

Course content

Monday, September 7, 2026 9:55 AM

1. 30 theoretical 70 hand on
2. Certification complete knowledge on terraform. Multiple choice question. Tru and false. Documentation
3. Dumps

How good you are with aws. 1 to 10

We will create network vpc subnet ig custom rtb attach ig with custm rtb launc h instance in private and public subnet. Using aws console
Then how to convert the same into terraform.

Region

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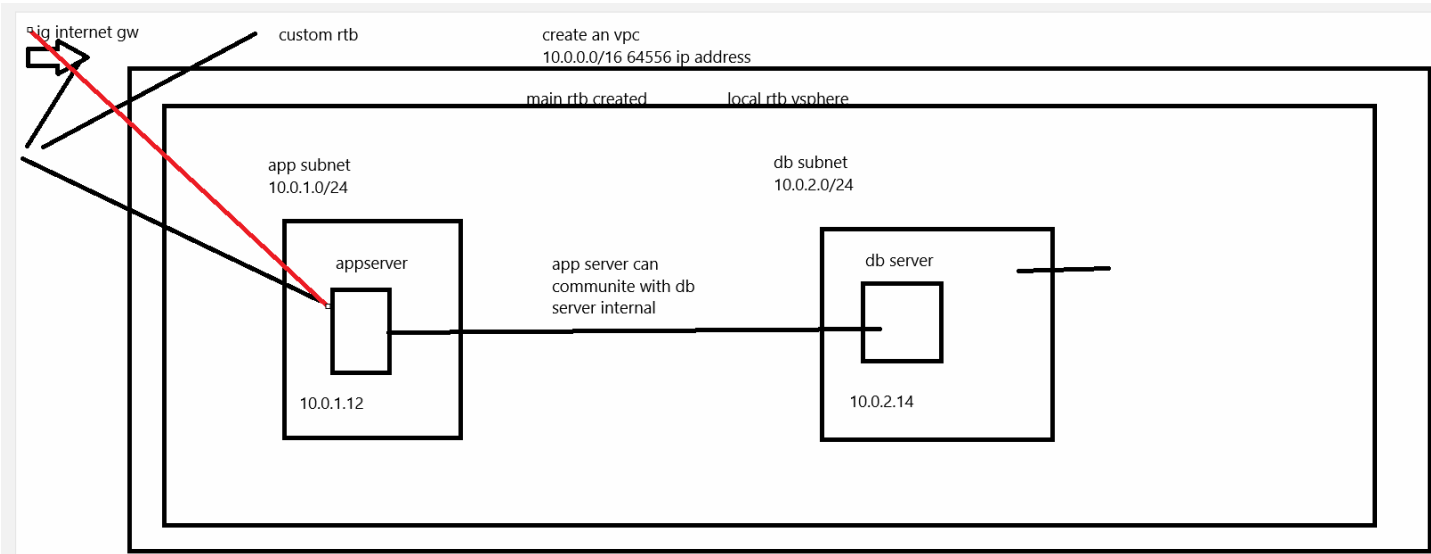
What is the purpose of this region?

1. Compliance
2. Latency

You migrate to a new location

You required a new internet connection

You also need to create an internal network.



app server wa
communicate
network.
or someone v
connection fr
your app serv

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IAC

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Why you need Infrastructure as Code?

Vmware platform you are running on prem. There is no overhead cost. Hardware cost is not there. No rental.

It keep on growing I don't need to maintain it also.

Cloud is all about rental. There is no capacity issue in rental.

5 years. 10k application. Flipkar phone pay

Culture.

Developer need to deploy frequently. Feature need to be tested. If you run it in a vm that way we will get 100% result.

Own data center cannot support. Openstack.

They tie up with azure and gcp.

They can deploy the app in the cloud. Test it and destroy it.

I want that env easily available. 30 sec once I finish my test cases I destroy it billing stop.

Constrain of resources in the cloud.

I need to control also

I don't want that my developer should login to the cloud and launch instance zero touch deployment.

I need to create template for different depratement to create the infrastructre. Dev t3.medium 2 core processor and 8 GB of ram.

Prod 4 core with 32 gb of ram.

1. The cloud infra creation will be done using code which is as per mu organization requirement policy
2. It should be cost effective.
3. With a single command we can destroy the resources.

Aws use cloud formation template

Azure arm template

Gcp has json terraform template.

Terraform has single language to communicate with all the cloud service provider

Stability

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Using terraform we build the infrastructure. Ec2 vpc storage 80GB-db guy change the storage from 80Gb till 120GB.
This is called as drift
Terraform apply it will show me the changes what is being done outside terraform.

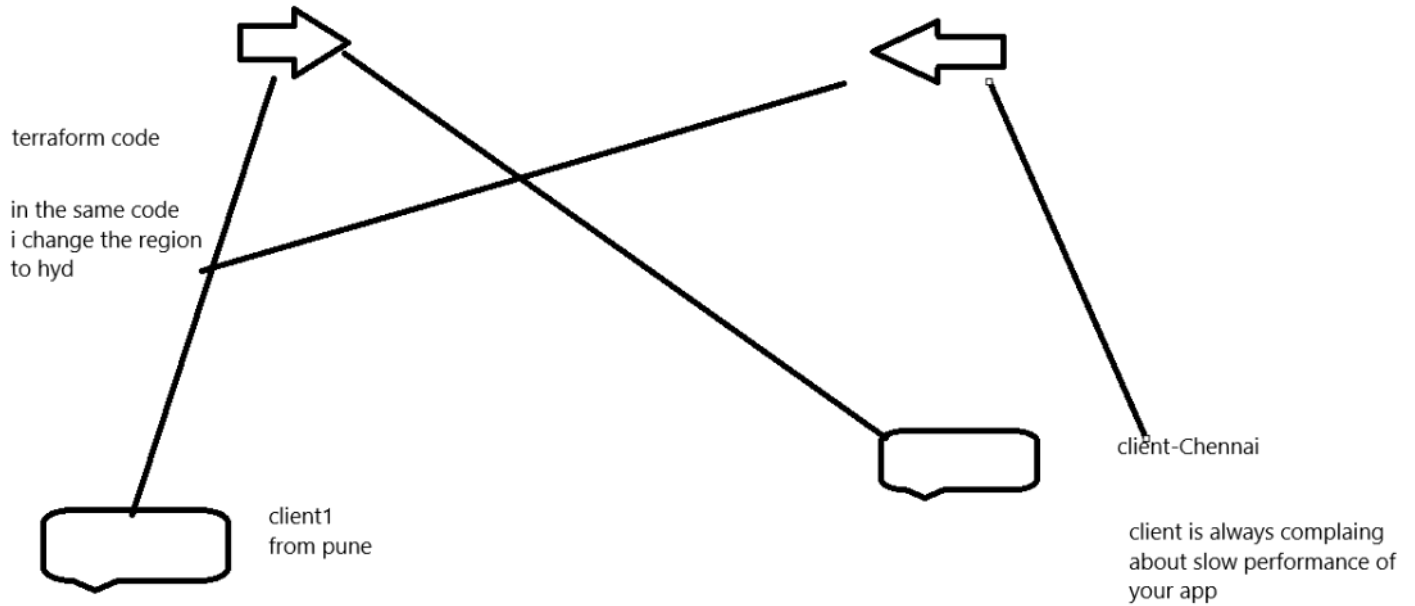
If you are managing your resources using terraform. You need to make that changes using terraform.

ecommerce app

streaming app

mumbai

hyderabad



Screen clipping taken: 9/7/2026 12:54 PM

I am not creating the documentation terraform machine is generating it.

Security

Monday, September 7, 2026 1:00 PM

Who created user in aws cloud

I will create a terraform template to create user in aws.

IAM team.

Upload this in version github. Github access to only the required teams

A new team member has join your org in your team only. He has access to all terraform code.

T2.large 4 core with 32 gb ram.

T3 xlarge 8 core 64 gb of ram.

Sentinel policy. Opa policy. Version control I ll apply this policy. Rego language

T2.large can be created

Terraform installation

Monday, September 7, 2026 2:20 PM

We will install a windows machine in aws.

Rdp

And install

Terraform

Git

Vs code

Aws cli.

To access this windows machine you need to disable global protect vpn pcap

We install all the necessary package

also authentication with the cloud service provider using access key and secret access key

Terraform workflow

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--->first we will create a directory-inside the directory we will put all the code-to create vpc-subnet customrtb-internetgw-attach ig with customrtb-subnetassociation-vmcreation

How to terraform need to know this is the current directory which is going to use for creating specific resources

Plugin:

In your vmware platform you want to connect an usb device. You need a plugin.

Aws use cloud formation template

Azure use arm template

Plugin is to convert your terraform code to aws understanding code.

Terraform plan: it will read all your terraform manifest file. Then compare it with cloud service provider and give you a dry run.

It will again read your files in the current dir

Terraform apply: will read all your manifest file and give you an desired state when I say yes it will convert the current state into desired state. Apply those changes in the cloud. All those changes need to be stored inside a file called as terraform.tfstate

Terraform destroy it will destroy all the resources. Which file it need to read to destroy the resources.

Vpc-resource

Argument-name and cidr

Using terraform I am creating an vpc

Argument and value against argument

Name:gopal-vpc

CIDR:10.0.0.0/16

Blocks

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Terraform block: here we will define which version of the terraform we will use to create the resources

Provider block: aws

Resources block: vpc subnet ig or I want to create an rds instance ec2 instance

Input variables:

I have an ecommerce application merchandize database. Running in indian market.

This db is running on ec2 instance-custom vpc subnet natgw sg. I found that it is perfectly no valnurability. Mumbai region.

I want to expand my market -- uae.

If I have provider

Region=var.region ---mumbai

---uae(dubai)

Terraform apply

I want to deploy my app in dev test prod

Environment=dev test

Output: using terraform I create the resources in the cloud. I create an use iam validate the user created using aws console

Display the usercreated in my terraform console only.

Local block: used for concation or muyltiple values mapping.

Data source block

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Calling reference block.

Data source block.

Anything you have created outside terraform we can call

1. Ec2 instance. We selected an ami.
2. Secret manager

Recap

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IAC: how IAC can help me to manage my resource in the cloud.

It when you need to build the resource you build it and when you want to destroy it you destroy the resources in the cloud.

Terraform workflow.

Terraform init: this will initialize your terraform directory contains all the terraform configuration files.

And as per the provider block it will download the provider plugin

Terraform validate: this will validate my code syntax validation it will do it.

Terraform plan: this is going to read all your manifest file in the current directory check the desired state as per the current directory and give you the desired state in a dry run

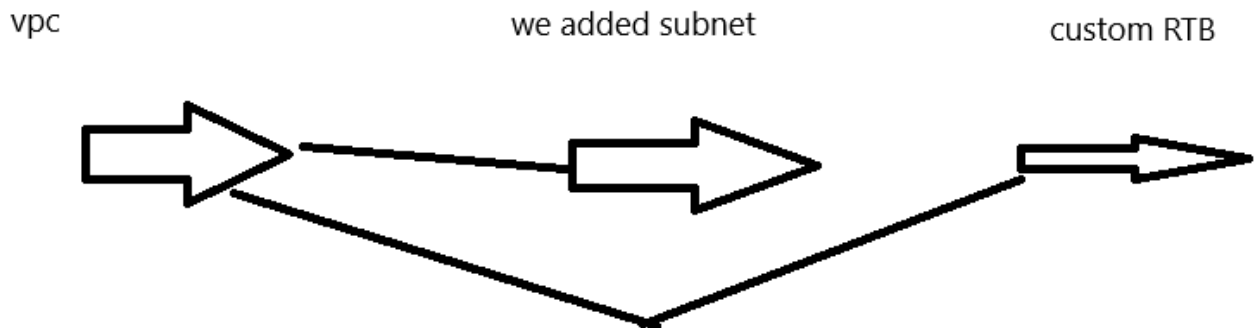
Terraform apply. It will create the resources in the cloud. Once the resource gets created it will store those resources inside a file called as tfstate.

Terraform destroy: this is going to destroy the resource in your provider which is the cloud reading your tfstate file.

Terraform reads only those files with the extension of .tf

We learn about different blocks in terraform

1. To create resource we create a block called as resource block.
2. We learn different kinds of variables: input variables, output variables, and local block.
3. Then to reference or call a block. Anything which is created outside terraform and you want to call it, it will be on a reference block.



how using terraform code we are going to map these resources.

Screen clipping taken: 9/8/2026 10:14 AM

Later on when we add vpc-subnet

Terraform apply

It should read my state file where already the vpc information is there. So it should not recreate the vpc only it should create the subnet.

Terraform is not an programming language

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Syntax based language

All this are being converted into go lang

But it is simplified language

Output block do: once we created the vpc and subnet. How to verify the same.

Terraform apply : how to confirm that vpc and subnet is created. . We can go to aws console and check the same. Or you can verify the same from tf state file.

We learn about three different block

1. Resource block: to create resources
2. Input variables: to
3. Output block. : to display values in the terraform console.

Input variables= we declare inside a tf file and do the calling using var.nameofthevariables

Output block: we call it from terraform.tfstate file

Input variables type called as object string will accept single value.

Input variables object

Tuesday, September 8, 2026 12:31 PM

What is az in aws. Availability zone

Inside a region in aws how to make my environment ha

Inside your vmware you deploy an application inside a vm. It is running on a physical rack. If the rack goes down

It should be deployment in two different rack.

Three month back aws one of the region goes down. Nv location down snapchat application is down.

Walmart also run some of the application in n.v that also goes down. In cloud how we can reduce the down time.

How you can make your retail chain the customer choice.

Your application should not be down anyhow at midnight I want to order some food and zomato down.

Resilient architecture.

Input variable type string can take single value

We want to create subnet this subnet need to be created in two different az

When we create instance similar kind of instance need to be created in two different az.

Dictionary.

Suppose I am maintain a dictionary of student in classroom

Student=student1-name=rahul age=20 city=delhi

Student

Against a variable I want to define multiple values

Should be in key value format

Object

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Every resource block can create a single resources

```
resource "aws_subnet" "web_subnet" {  
  vpc_id      = aws_vpc.my_vpc.id #it will read it from state file  
  cidr_block = var.subnet_cidr  
  Az -az-1  
  tags = {  
    Name = var.subnet_name  
  }  
}
```

#to create multiple subnet if we create the resource block multiple times

##it will be like repeation of the code

#it will make my code confusing

#not an readable

When you want to do similar kind of job in any programming language

What you need to do .

Loops

Now terraform is not an programming language.

Loops are called as meta argument

College db

Student db

Select from student where student name phone email id

Local block

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Named expression

Gopal

#expression should be always in key value format

Happiness: Raised Cheeks

Sadness: loose eyelids

Anger: tensed eyelids

Surprise: lifted eyelids

If I call gopal

I will get all the three expression

This is called as named expression against gopal

Call gopal

In terraform when you want to create named expression you need to use a block called as local block

1. You can use the same for named expression
2. Concatenate your variables together

Vpc

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We will create a custom rtb attach it with the vpc

Ig attach the ig with my vpc

Finally map my subnet to the custom rtb

#internet gw and attach with the vpc

Recap

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Using this pattern we can create private subnet without any internet connection.

Already you have the resource available to create this private subnet

Object variables

Route table

Data block

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We learn about

Resource block: this will create resource

Input variables: pass input variables

Output block: display output in the terminal

#we are going to learn about a calling block called as data block.

#what you need to call using terraform

This image is created outside terraform. When you try to create an instance you want to call an image which is created outside terraform. You can use a block called as data block

Where do you store your secrets. Api key or password storage or s3 bucket or vault secret manager.
While creating the instance our application want to call secret

Vm

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We will create an startup script boot strap script.
Linux machine we need to give it in shell format
Windows machine we need to give the same in ps1 format
Cloud init

HA

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To create ha environment we have created three subnet. And this subnet are in three az data center.

We have two challenges

1. I have three az I want to create three instances in this three az but I also want every az the size of the instance should be different.
2. Az : t2 micro az2: t3.micro az3:t2.medium
3. All the three az will have similar kind of instance

How to achieve this .

Three different in all the az the instance size will be different.

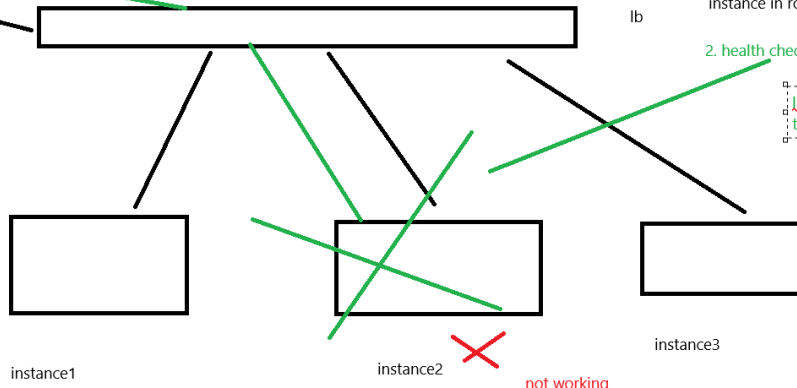
Object

We need to create multiple instance in multiple az also we want to create those instance with different configuration deployed in az

So when I want to pass multiple values we have variables option as object variables.

We are accessing all the three instances using individual ip address

acme.com



1. first job of lb will be to distribute the traffic to multiple instance in round robin fashion

2. health checkup

lb will not send the traffic to that particular instance

i am running an ecommerce application in this instance. we will provide individual ip address to the customer

Lb

Wednesday, September 9, 2026

3:06 PM

1. Name
2. Internet facing
3. Select vpc and subnet
4. Attach a sg
5. Listener and routing.

Target group

1. Name of the tg
2. Select the vpc
3. Health checkup

recap

Thursday, September 10, 2026 9:47 AM

We created an entire resource upto load balancer.
How do we refer or make an resource part of another resource
Vpc-inside the vpc we are adding subnet.

We used reference block. Using the same only we have mapped

We have use a new block called as data block in terraform. If you create anything outside terraform you are going to call the same using data block.

Remote state

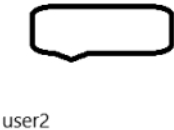
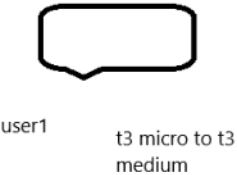
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When you want to store your file in a remote place.
State file is very much important

you are storing the state file in your local system
there is no versioning available
terraform apply or destroy
1. terraform apply: it will create the state file
2. terraform destroy it will take the previous state and store in a file terraform.tfstate.backup
because of read write nature of your storage.

on the day 1 we create vpc and subnet
destroy
day 2 we created vpc subnet ig
so once you destor the day 1 i lost my history

and suddenly the auditor comes all the history should be
avaialble



both the user with the same code apply the changes from
there workstation what will happen
it dont conflict if your state file is in local machine it will try
to create the resources which is like an disaster

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Object storage

Thursday, September 10, 2026 10:19 AM

You need an storage service from the cloud where if I upload the same file multiple times then also it should not overwrite

Cloud service provider has an storage s3 object storage you have the option of enabling versioning. If you upload the same file multiple times using the timestamp they are versioing the file

Terraform cloud

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Terraform cloud cannot store your code.

To store your code you need to integrate terraform cloud with a version control system like git.

Git

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Working

How using the git we can manage the code.

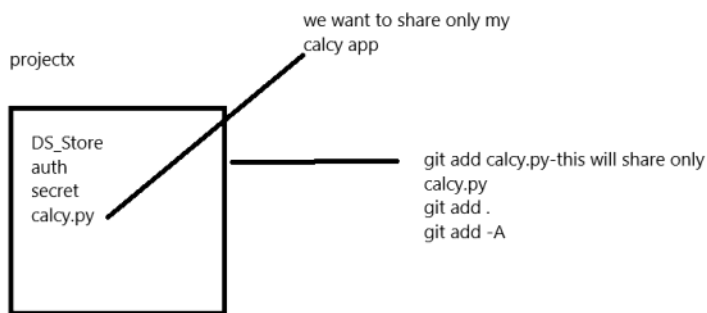
Now you are coding. You need to also know how to manage the code. Get an history what changes you have done when you have done.

Git global variables

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User.name email id :

git init



git init
git init will create a folder called as .git
this project folder in git terminology we called it git working directory

you need to add addition function in calcy application . you modify your code add the function
git add calcy.py

git add will move your calcy file inside a area called as staging area

git commit -m "working on calcy project"

whatever file we have in staging area will be committed

git commit -m "added a new function"

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Git remote

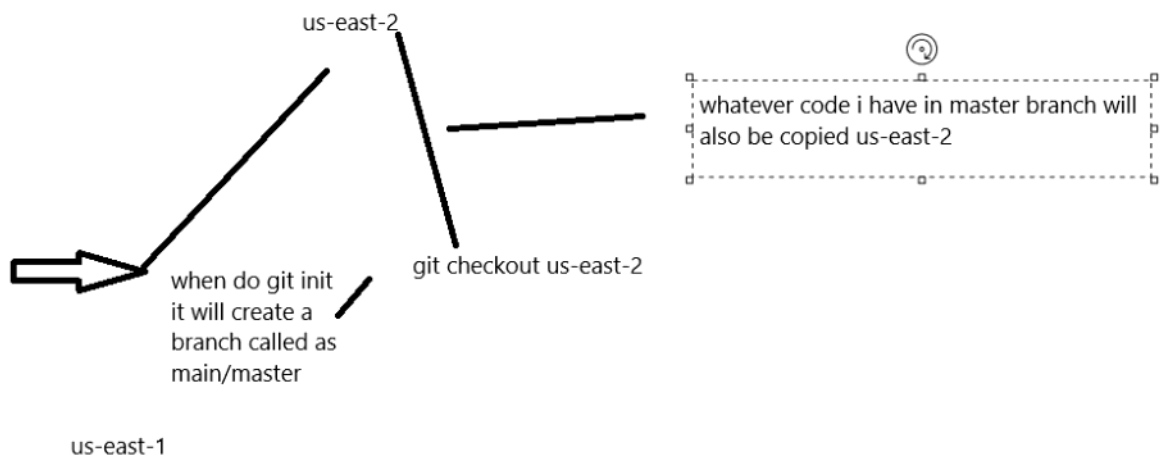
Thursday, September 10, 2026 12:46 PM

Once you make changes in the code which is requested by the client you do a commit on those changes. Those changes are there in your local machine.
Share the same with the client .
We need to upload my code in a remote repository .
Github azurerepo bitbucket gitlab.

Git branches

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Using the same terraform code we want to deploy similar kind of infra in different region



Screen clipping taken: 9/10/2026 3:03 PM

Using the same code I can create dev test prod environment or may be I want to change with the region Without disturbing the main code I can copy all the code in a new branch and start updating the same.

Git and gitbranch

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Git to manage our code

When we want to extend our git functionality we use git branching.

code

Thursday, September 10, 2026 3:27 PM

1. We have created terraform code I need to upload the same in github.

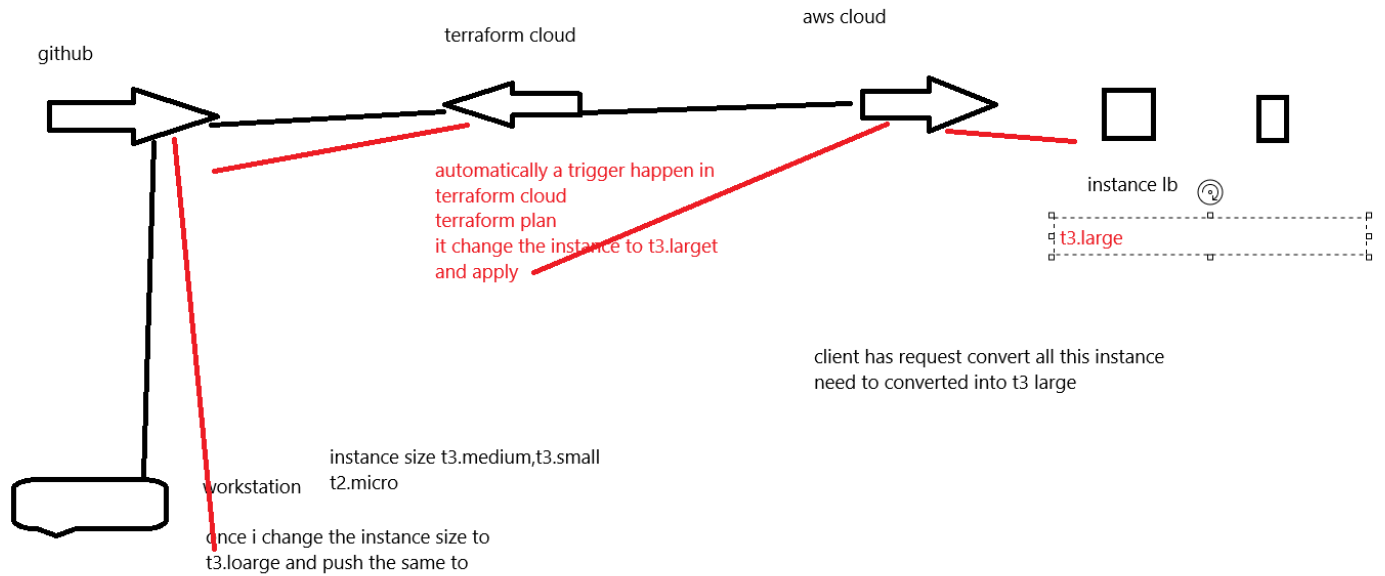
Recap

Friday, September 11, 2026 9:47 AM

1. We are left out with terraform cloud
2. We have already done the integration of terraform cloud with github
3. We also provide the auth using aws access key and secret access key
4. Need to auto-run the infrastructure. Enable the auto-run settings
5. We also enable the cost factor

Cloud integration

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Screen clipping taken: 9/11/2026 11:03 AM

Once the instance size change

1. It should destroy and recreate
2. But in cloud it will reboot the instance and change the size

Variables files

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We discussed about git branching.

Similar to that terraform has something called as workspace. It is kind of branching only

To create multiple environment using terraform dev test prod

##terraform has something called as workspace

When you use terraform workspace we need to pass on the variables file also dev.tfvars prod.tfvars. All the variables declaration values.

Modules

Friday, September 11, 2026 2:15 PM

You need to make your code more reusable.

How to make my code reusable.

Module will help you to do the same.

We have the code and we pass the variables to the code. Terraform apply we can create the resources.

What are the common task I have which can be clubbed together. That way only we can build an terraform modules

Create a line where body is being created

Another where the painting is done.

Finally in the last line we attach everything.

Networking

Security

Compute

Lb

I work closely with walmart.

Technology

Every month they hire more than 30 people.

They goes into multiple department.

Every month creating this 30 user in azure entra id. Active dr.

HIPAA or PCI DSS.

Policy need to maintainer.

Using terraform we created user wise department wise module.

Username and department. Terraform code select particular department and put the user id. Terraform apply.